PRANAV JAIN

University of Southern California

Email: pranavj@usc.edu

[Website](), [Scholar]()

RESEARCH INTERESTS

Geometry Processing, Physical Simulations, Computer Graphics, Numerical Analysis of Partial Differential Equations, Discrete Differential Geometry

EDUCATION

University of Southern California (USC) — August 2023 - Present
Doctor of Philosophy (PhD), Computer Science — *Los Angeles, USA*

New York University (NYU) — September 2021 - May 2023
Master of Science (MS), Scientific Computing — *New York, USA*

Indraprastha Institute of Information & Technology Delhi (IIITD) — August 2016 - August 2020
Bachelor of Technology with Honors (B-Tech Hons), — *New Delhi, India*
Computer Science and Applied Mathematics

PUBLICATIONS

Pranav Jain, Navami Kairanda, Peter Yichen Chen, & Oded Stein. (2026). PINNsur: Physics-Informed Neural Networks for PDEs on Curved Surfaces. [link]

Pranav Jain, Ziyin Qu, Peter Yichen Chen, and Oded Stein. 2024. Neural Monte Carlo Fluid Simulation. In ACM SIGGRAPH 2024 Conference Papers (SIGGRAPH '24). [link]

Liam Martin, **Pranav Jain**, Zachary Ferguson, Torkan Gholamalizadeh, Faezeh Moshfeghifar, Kenny Erleben, Daniele Panozzo, Steven Abramowitch, Teseo Schneider. A systematic comparison between FEBio and PolyFEM for biomechanical systems. In Computer Methods and Programs in Biomedicine, 2023. [link]

Zachary Ferguson, **Pranav Jain**, Denis Zorin, Teseo Schneider, and Daniele Panozzo. High-Order Incremental Potential Contact for Elastodynamic Simulation on Curved Meshes. In ACM SIGGRAPH 2023 Conference Proceedings (SIGGRAPH '23). [link]

Pranav Jain, Munawar Hasan, Donghoon Chang. Spy based analysis of selfish mining attack on multi-stage blockchain. In Cryptology ePrint Archive, 2019. [link]

RESEARCH EXPERIENCE

Google Summer of Code [link] — May 2026 - August 2026
Contributor — *Remote*
Advisor: Prof. Martin Skrodzki, Dr. Andreas Fabri

- Implemented the algorithm proposed in *Lipschütz et al. (2024). Manifold Meshing of Point Clouds with Guaranteed Smallest Edge Length* for mesh reconstruction from scratch and integrated it into the Computational Geometry Algorithms (CGAL) library

Adobe Research — May 2025 - August 2025
Research Intern — *Seattle, USA*
Advisor: Dr. Timothy Langlois, Dr. Danny Kaufman, Dr. Jui-hsien Wang

- Researched on how to make generative-AI models more physics accurate by limiting hallucinations for video generation

University of Southern California August 2023 - Present
PhD Student *Los Angeles, USA*
Advisor: Prof. Oded Stein, Prof. Krishna Garikipati

- Researching on empirically proving convergence rates of Poisson problem using Physics Informed Neural Networks (PINN) with respect to the size of the architecture
- Researched and published on a discretization-free fluid simulation method using combination of Monte-Carlo and neural fields

New York University September 2021 - May 2023
Research Assistant *New York, USA*
Advisor: Prof. Daniele Panozzo, Prof. Denis Zorin

- Researched and compared the simulation abilities of PolyFEM and FEBio in terms of accuracy, time and user inputs
- Introduced a high-order finite element formulation for elastodynamic simulation with contact handling

nTopology June 2022 - August 2022
Software Engineer Intern in Geometry Team *New York, USA*
Advisor: Suraj Musuvathy, Ranjeeth Mahankali

- Designed and implemented a method to efficiently stitch different CAD models to form one unified CAD structure

Freie Universität Berlin [link] September 2020 - August 2021
Research Intern *(Virtual) Berlin, Germany*
Advisor: Prof. Konrad Polthier, Dr. Sunil Kumar Yadav

- Researched on a robust and parameter-tuning free method to denoise a point cloud

Fields Undergraduate Summer Research Programme 2020 [link] July 2020 - August 2020
Research Intern *(Virtual) Toronto, Canada*
Advisor: Prof. Thomas Uchida

- Researched on the problem of inverse mechanism synthesis – given a path, design a mechanism that traces the path

Indraprastha Institute of Information & Technology Delhi [link] August 2018 - August 2020
Research Assistant *New Delhi, India*
Advisor: Prof. Kaushik Kalyanaraman

- Developed DECAGT, a software framework for discretizations of the objects and operators of exterior calculus

Indraprastha Institute of Information & Technology Delhi [link] August 2018 - August 2020
Undergraduate Thesis *New Delhi, India*
Advisor: Prof. Donghoon Chang

- Proved the selfish mining attack strategy on the Bitcoin blockchain for multiple mining pools
- Designed and proved an extended attack for multi-stage blockchain

ACADEMIC COMMUNITY WORK

Tertiary reviewer for SIGGRAPH 2026 February 2026

Tertiary reviewer for SIGGRAPH Asia 2025 and Pacific Graphics (PG) 2025 July 2025

Volunteered as a teaching assistant for the Poisson Reconstruction Project at Summer Geometry Initiative (SGI) 2024 July 2024 - August 2024

Member of the Conference Coffee team at SIGGRAPH Research Career Development Committee. 2021-2022

TEACHING EXPERIENCE

University of Southern California

Teaching Assistant: CSCI 532 - Innovation for Defense Applications Spring 2026
Teaching Assistant: CSCI 103 - Introduction to Programming Fall 2025
Teaching Assistant: CSCI 580 - 3D Graphics and Rendering Spring 2025
Teaching Assistant: CSCI 596 - Scientific Computing and Visualization Fall 2024
Teaching Assistant: CSCI 104 - Data Structures and Object Oriented Design Spring 2024

New York University

Grader: MATH 263.3 - Applied Partial Differential Equations Spring 2023
Grader: MATH 252 - Numerical Analysis Spring 2023
Grader: MATH 263 - Partial Differential Equations Fall 2021